

Replacing Batteries with a Substitute in a Solar System

Template

Some of the information shown on the template are different and some are identical to the template for the GA. The documents provided must be different although it can be similar. If you copy from yourself then it is not considered plagiarism. This template can just be the GA template changed to accommodate everything required here. The part that follows is the template for the record you will need to submit. Read through this template and replace all the statements and questions in green with your own statements. Your statements must address the questions and statements made in the template. Thus, the questions, statements and drawings are in the template to guide you. Delete all the statements in green. Thus, do not keep the questions. Answer the questions as if though it was just a thought that you came up with. It is recommended that you keep the headings.

Cover or Title page

Name of the practical

Subject

Name of student

Date practical was performed

Date report was handed in

Declaration that the report is the author's own work

Signature

This page must not be numbered

Abstract

It is recommended that you do this part last. This does not mean that you should put it last in the document. Thus, the abstract must follow the cover page. This part has to contain an overall of everything. This is why it is recommended to do the abstract after the rest of the document is finished. Look at each section in the rest of the document. In one or two sentences, explain what happens in each section. Do this with all the sections starting with the introduction and ending with the conclusion.

Terms of reference

Table of contents

List of Illustrations

Indicate which pictures are on which pages. Pictures that you will need to include is of the simulation and the physical layout.

Glossary of terms

Explain what different abbreviations, which you are going to use, means. For example: D.C means Direct Current.

Introduction

Subject

Background

Problem statement

Objectives

Explain what you are trying to achieve in the end.

Methodology

Explain which steps you have taken in the designing process.

Project and Data Design

Circuit diagrams (20% of FISA)

You have to draw the circuit diagram from scratch. If you do not draw the circuit diagram from scratch, then marks will be deducted for you. Thus, do not just download the circuit diagram from TinkerCad.

Calculations (5% of FISA)

The calculations has mostly to do with calculating the values of resistors and capacitors. Some resistor and capacitor values are given. Thus, you have to use these resistor and capacitor values to calculate the other resistor and capacitor values. Do not calculate resistor and capacitor values that were provided to you.

Photo of Simulated Circuit (10% of FISA)

Take a screenshot of the hardware simulation part of Tinkercad. Paste the screenshot into this position of the documentation.

Photo of Physical Circuit (10% of FISA)

You are required to take a photo of the physical layout of your “solar charge controller”. Make sure all the connections to all the components are clearly visible. Make sure your student card is visible in the picture as well. This picture must be added to this position in your documentation.

Program code (35% of FISA)

Copy and paste the code as text into this part. Do not insert pictures or screenshots of code. The program should be able to do the following:

- Charge the capacitors as far as possible.
- Switch capacitors from series to parallel and back to minimise losses while providing appropriate voltages to the load.
- Adjust output voltage using infrared remote. The output voltage must be sensed and this sensed voltage must be used to adjust the output voltage to be the required voltage.
- Show voltages on LCD display. These voltages should be calibrated. When the output voltage is changed by the user then the LCD screen should provide a display that will help the user insert a new value.
- Create custom characters to show when capacitors are in series and when in parallel.

Findings

Explanation of findings

Explain what was working and what was not working. Also explain how well did the circuit work. Include a screenshot of the serial monitor for the simulation as well as the physical circuit.

Rectifications based on findings

During the designing process, a person encounter's bugs and thing that does not work properly or at all. Indicate what was not working correctly. Then indicate what you have done to resolve this problem.

Conclusion

Scope and Limitations

Indicate what your project will not be able to do. Also, indicate the implication of this limitation.

Recommendation

If a person has to redesign your project, tell them what advice you can give the designer.

Appendices

This part is only needed if you have extra pages that you want to show to the reader.

References

Place all references here. Use Harvard referencing. Everything that you have read or looked at to make this possible has to be included over here.

Bibliography

Student name.....

Student number.....

Category	Description	Examiner	Moderator	Maximum mark	
Record FISA	Circuit Simulation Picture	Communication devices		1	10
		Voltmeters		1	
		Battery replacement		1	
		Relay		1	
		Protection		2	
		Solar panel		1	
		Voltage reading circuits		1	
		Filter		1	
		Regulator		1	
	Physical Circuit Picture	Communication devices		2	10
		Battery replacement		2	
		Relay		1	
		Protection		1	
		“Solar panel”		1	
		Voltage reading circuit		1	
		Filter		1	
		Regulator		1	
	Software	LCD displays		1	35
		Surname displayed on LCD		1	
		Custom characters shown		2	
		Output correct to two decimal values		2	
		LCD voltage corresponds to voltmeter readings		2	
		Switch from parallel to serial.		2	
		Voltmeter readings confirm parallel to serial switch.		2	
		Symbol changes from parallel to serial.		2	
		Switch from serial to parallel.		2	
		Voltmeter readings confirm serial to parallel.		2	

		Symbol changes from serial to parallel switch.			2	
		The output voltage seems to be fairly constant.			2	
		It seems the output voltage is regulated			2	
		Buttons on remote is registered.			2	
		Values are updated and shown			2	
		Two values are updated and shown			1	
		Return to normal screen after 1 second			2	
		Value can be changed multiple times			2	
		Output voltage changes to new value.			2	
	Record layout				1	20
	Circuit diagram	Drawn from scratch			3	
		Connections between relay and components			3	
		Protection			3	
		Voltage reading circuits			2	
		Voltage balancing component			1	
		filter			3	
		Voltage regulator			2	
		Load			1	
	Renewable energy				1	5
	Calculations	Output voltage			1	
		20V voltages			4	

Demonstration	Student was able to give condition for switch from parallel to series.			2	10
	Symbols confirm switch			1	
	Voltages confirm switch			1	
	Voltages on components similar but not identical			1	
	LCD voltage correspond to voltmeter readings			2	

	Student was able to adjust output voltage			2	
	Output voltage on voltmeter systematically changes to inserted value			1	
Presentation	Timing			1	10
	Readable			1	
	Logical			3	
	Pictures			1	
	Dressed formal			1	
	Answers			3	
Total				100	100